

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

19th May 2023

Subject CB2 – Business Economics

Time allowed: 3 Hours 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** Which of the following are complementary good?
- A. Car Insurance and Car
 - B. Tea and Coffee
 - C. Keyboard and LED Monitor
 - D. All of the Above
- [1.5]**
- Q. 2)** A subsidy for car production will lead to?
- A. Shifting of demand curve to the left
 - B. Shifting of demand curve to the right
 - C. Shifting of supply curve to the right
 - D. Shifting of supply curve to the left
- [1.5]**
- Q. 3)** A product's demand curve QD is $100-2P$. Calculate the Price elasticity (point elasticity) of demand at price of INR 10.
- A. -0.125
 - B. 0.25
 - C. -0.25
 - D. 8
- [1.5]**
- Q. 4)** The Cross-price elasticity of demand is 1.20 for Good X with respect to Good Y. Which of the following is true?
- A. Good X and Good Y are likely to be close substitutes
 - B. Increase in Price of Good Y will lead to a proportionate increase in price of Good X
 - C. Good X and Good Y are complementary goods
 - D. None of the Above
- [1.5]**
- Q. 5)** Shadow markets are most likely to be associated with
- A. Price ceilings which cause excess demand
 - B. Price ceilings which cause excess supply
 - C. Price floors which cause excess demand
 - D. Price floors which cause excess supply
- [1.5]**
- Q. 6)** Which is not true about a risk averse investor?
- A. He or she will reject a fair gamble
 - B. He or she will avoid all risk
 - C. The marginal utility of wealth decreases as wealth increases
 - D. The insurance premium can be greater than the long run average value of claims
- [1.5]**
- Q. 7)** Which of the following factors of production are variable?
- I. Raw Materials
 - II. Electricity
 - III. Manager employed on a contract
- A. I and II
 - B. I, II and III
 - C. I only

D. II only [1.5]

Q. 8) In a perfectly competitive market

- A. The Products are identical
- B. The consumers have perfect knowledge and behave rationally so always buy the cheapest good
- C. Demand is perfectly in-elastic
- D. Both A and B

[1.5]

Q. 9) Which of the following is second degree Price discrimination?

- A. A street vendor of fruit charging a maximum price that the customer wants to pay
- B. An Insurance company grouping potential customers in segments of high net worth, medium and mass market and charging them differential rates
- C. A supermarket charging different price for different products
- D. A wholesaler of flowers charging lower rates from a consumer buying in higher quantities

[1.5]

Q. 10) Which of the following is NOT a social cost of plastic industry?

- A. Cost of treating illness from consumption of plastic in food
- B. Environmental hazard caused by microplastics in the environment
- C. Both A and B
- D. None of the Above

[1.5]

Q. 11) Which of the following was primary reason Keynes came up with his theory known as Keynesian school of thought?

- A. The classical approach couldn't explain the persistent unemployment during the great recession.
- B. Free market theory couldn't explain the inequalities between rich and the poor.
- C. The classical theory of free markets would bring about maximum social benefit.
- D. The classical approach suggested that intervention by the state would reduce nations' wealth.

[1.5]

Q. 12) Which of the following is TRUE for New Classical school of thought?

- A. Government management of demand
- B. Targets for money supply growth
- C. Surplus value
- D. Rational expectations

[1.5]

Q. 13) The marginal utility for an individual is measured as the maximum amount (in Rs.) that they are willing to pay for consuming a unit of Good A.

The table below shows the marginal utility for consuming various units of Good A priced at Rs. 12 per unit.

Quantity consumed	0	1	2	3	4	5
Marginal Utility (Rs.)		30	25	20	15	11

Individual's marginal consumer surplus from consuming a third unit of Good A is:

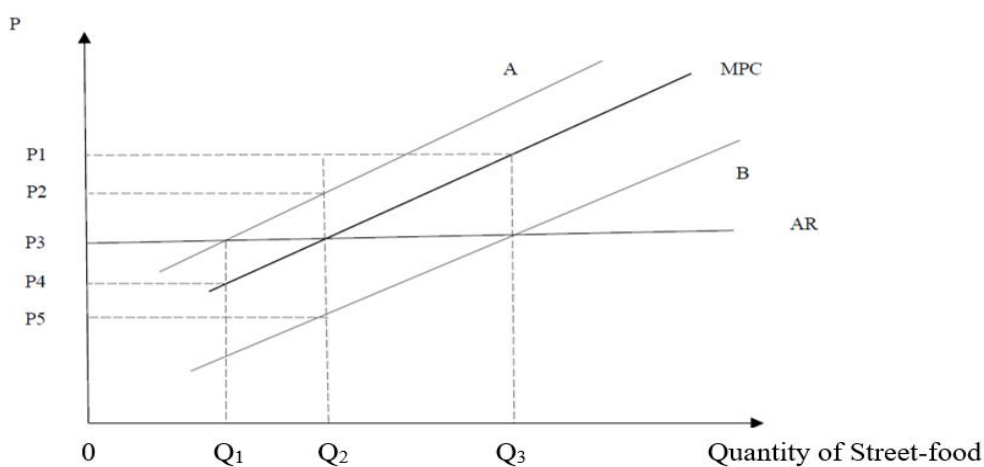
- A. Rs. 16
- B. Rs. 39
- C. Rs. 8
- D. Rs. 24

[1.5]

Q. 14) Assume that the production of the street- food leads to pollution of the surrounding land area, but this pollution is an external cost to the street- food vendor.

Following diagram represents a profit-maximising street-food vendor operating in perfect competition, where MPC is the Marginal Private Cost curve and AR is the Average Revenue curve for the vendor.

Note: P denotes Price.



If the vendor decides not to take any account of the damage to the surrounding land area, the level of output the vendor will produce and the corresponding marginal external cost are given by:

- A. $Q_1, P_3 - P_4$
- B. $Q_2, P_2 - P_3$
- C. $Q_2, P_3 - P_5$
- D. $Q_3, P_1 - P_3$

[1.5]

Q. 15) The Pareto optimum level of output is the level of output at which

- A. It is impossible to make anyone better off without making someone else worse off
- B. This occurs where Marginal Social Benefit (MSB) is greater than Marginal Social Cost (MSC)
- C. The MSB is equal to marginal private benefit plus the marginal external benefit
- D. A and C are true

[1.5]

Q. 16) Which of the following should decrease the money supply?

- A. RBI buys Government bonds from the market
- B. RBI sends a circular to all major banks to reduce their cash reserve ratio
- C. RBI sells Government bonds in the market
- D. Through the voluntary disclosure scheme the Government extracts from the private sector an additional tax revenue of Rs. 5000 crores

[1.5]

- Q. 17)** Which of the following does not work as an Automatic stabilizer to pull an economy back from recession?
- A. High PSNCR
 - B. Low interest rate
 - C. Low capital investment
 - D. Stock building
- [1.5]**
- Q. 18)** Suppose the Indian Rupee is undervalued, then
- A. India is hurting exporters
 - B. India will have a balance of payments deficit
 - C. India will accumulate reserves of foreign currency
 - D. India will experience downward pressure on its exchange rate
- [1.5]**
- Q. 19)** Assume that the Government imposes minimum wages regulations and bans child labour. This would lead to:
- A. Demand-pull inflation
 - B. Cost-push inflation
 - C. Both
 - D. Neither
- [1.5]**
- Q. 20)** Consider a simple Keynesian model for an open economy. Suppose that a ban is imposed on imports. Following this, GDP
- A. Will decline
 - B. Will increase
 - C. Will remain unchanged
 - D. Nothing can be predicted without knowing the nature of the import function
- [1.5]**
- Q. 21)** An increase in CRR (Cash Reserve Ratio)
- A. squeezes credit supply and keeps money supply unchanged
 - B. squeezes money supply and keeps credit supply unchanged
 - C. squeezes credit supply and may keep money supply unchanged
 - D. squeezes credit supply by the commercial banks and also money supply
- [1.5]**
- Q. 22)** One possible explanation for involuntary unemployment is:
- A. The wage rate does not adjust immediately to changes in the labour demand.
 - B. The labour market is perfectly competitive.
 - C. The firms, knowing the supply and demand curve for labour, implement the equilibrium wage, which happens to be so low that many people prefer to stay at home.
 - D. Maximum wages which are implemented.
- [1.5]**
- Q. 23)** Select the correct statement/statements which are correct as per the famous ‘Say’s Law’ using the code given below:
- I. Economic systems are ‘supply-led’
 - II. All accruing income is to be spent
 - III. Aggregate supply creates its own aggregate demand in an economy

- A. All of the above
 - B. II and III
 - C. I and II
 - D. I and III
- [1.5]**

Q. 24) In the context of balance of payments, 'portfolio investment' relates to:

- A. Takeover of foreign companies
 - B. Purchase of shares in foreign companies
 - C. Purchase of real estate in foreign companies
 - D. Private remittances to foreign countries
- [1.5]**

Q. 25) Consider the below statements:

- I. Saving is an injection and Investment is a leakage in the circular flow of income
- II. Imports are a leakage and exports are an injection in the circular flow of income

Which of the below is correct?

- A. I is false and II is true
 - B. Both I and II are false
 - C. I is true and II is false
 - D. Both I and II are true
- [1.5]**

Q. 26) The recession phase of the trade cycle

- A. Is often caused by excessive consumer expenditure
 - B. Is normally characterised by accelerating inflation
 - C. Is most prolonged when the country has high levels of imports
 - D. Is usually caused by falling aggregate monetary demand
- [1.5]**

Q. 27) Discuss the different ways in which the Government can intervene in the free market for welfare of the consumers and society in general and its relative merits. **[5]**

Q. 28) The Demand and supply conditions in a market of cloves are given by the following equations:

$$Q_d = 3600 - 5P$$

$$Q_s = 7P$$

Where P is price in INR per kg

The Quantity is in kg

- i) The Government is thinking about implementing a tax on the output of cloves to have some money for health insurance of clove farmers. What would be the new equation of the supply curve if a tax of INR 5000 per quintal is introduced? (1)
- ii) Due to introduction in tax will the equilibrium quantity demanded reduce? If yes then by what percentage? (3)
- iii) What would be the Tax revenue for the Government? (1)

iv) What would be new equilibrium price after tax? What is the share of producer and the consumer? (2)

v) Draw a diagram (Demand Supply Curve) explaining the impact of tax on demand curve and supply curve. (3)

[10]

Q. 29) A pandemic has struck the nation. For a few essential items like hand sanitiser the Government has imposed a price ceiling so as to protect the customers. Supporting your comments with appropriate diagram, discuss the advantages and disadvantages of this decision/ initiative. **[7]**

Q. 30) Your friend who has just started to read about economics is confused why rice is so much cheap as compared to diamonds whereas rice being a food item is essential for survival. Draft a letter explaining it to him in terms of Marginal Utility Theory. **[5]**

Q. 31) For each of the following explain whether it's perfectly contestable market

i) Housing Market (building flats for example) (2)

ii) Grocery store (2)

iii) Airline Industry (2)

[6]

Q. 32) What is the proposed by Neo-Keynesian school of thought and what are its key drawbacks? **[4]**

Q. 33) Assume prices and interest rates are fixed. An economy is described by the following equations:

$$\text{Consumption} = 100 + 0.6 Y_d$$

$$\text{Investment} = 100$$

$$\text{Government spending} = 200$$

$$\text{Export} = 40$$

$$\text{Import} = 40 + 0.1 Y$$

Where, Y = National Income

The rate of taxation is 50% of all income.

Calculate:

i) Equilibrium level of GDP. (3)

ii) Equilibrium level of consumption expenditure. (1)

iii) Trade deficit (1)

[5]

- Q. 34)** Define the following:
- i)** Hysteresis (2)
 - ii)** Structural Deficit (1)
 - iii)** Accelerator Coefficient (2)
 - iv)** J curve effect (1)
- [6]**
- Q. 35)** Assuming both India and US are allowed to float freely, discuss the factors that determine the exchange rate for Indian Rupees against the US dollar in the short term and in the long term. **[8]**
- Q. 36)** Define the bank deposits multiplier & the money multiplier and explain why the latter typically has a smaller value than the former. **[5]**
